

1. AI-Powered E-Commerce Platform

Role: Software Development Manager / AI Engineer

URL: [Universal Statues LLC | US Leading Fiberglass Statues Provider | Universal Statues LLC](#)

Overview

Designed and developed an AI-powered e-commerce platform for Universal Statues that combines customer support, marketing automation, and AI-generated content into a unified cloud-native ecosystem.

The platform consists of multiple serverless AI microservices responsible for customer interactions, automated marketing campaigns, social media content generation, and knowledge retrieval.

Business Problem

The company wanted to improve customer engagement while reducing the manual effort required for customer support, marketing campaigns, and social media content creation.

Solution

Designed the overall AI architecture using FastAPI microservices deployed on AWS Lambda. Built a context-aware AI chatbot using LangChain and LangGraph, implemented Retrieval-Augmented Generation (RAG) using MongoDB Atlas Vector Search, and integrated OpenAI and Google AI services for intelligent customer engagement and marketing automation.

Responsibilities

- Designed the overall cloud-native AI architecture.
- Built a context-aware AI chatbot using LangChain and LangGraph.
- Implemented RAG pipelines using MongoDB Atlas Vector Search.
- Developed FastAPI serverless microservices deployed on AWS Lambda.
- Monitored AI workflows using LangSmith.
- Integrated OpenAI and Google AI Studio.
- Designed reusable AI orchestration workflows.

Technologies

- Python
 - FastAPI
 - AWS Lambda
 - AWS SAM
 - API Gateway
 - LangChain
 - LangGraph
 - LangSmith
 - MongoDB Atlas Vector Search
 - OpenAI API
 - Google AI Studio
 - VoyageAI
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2. AI Marketing Automation Microservice

Overview

Designed and implemented an autonomous AI Marketing Agent that automatically creates personalized customer follow-up campaigns and integrates directly with Zoho Campaigns.

The service operates independently as an AWS Lambda microservice triggered by Amazon EventBridge Scheduler.

Business Problem

Marketing campaigns were created manually, making customer follow-ups time-consuming and inconsistent.

Solution

Developed an AI workflow that generates personalized HTML email campaigns using LLMs, automatically creates Zoho Campaigns, and schedules campaign delivery without manual intervention.

Responsibilities

- Built AI workflows for personalized email generation.
- Integrated directly with Zoho Campaigns API.

- Automated campaign creation and scheduling.
- Generated production-ready HTML emails using LLMs.
- Designed autonomous serverless microservices.
- Implemented scheduled execution using Amazon EventBridge.

Technologies

- FastAPI
 - OpenAI API
 - Zoho Campaign API
 - AWS Lambda
 - Amazon EventBridge
 - LangGraph
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3. AI Social Media Marketing Agent

Overview

Developed an AI-powered social media automation platform capable of generating promotional content and publishing posts automatically.

The platform retrieves product information through RAG pipelines before generating captions, marketing images, and promotional videos using Google Veo 3.

Business Problem

Creating social media content manually required significant marketing effort and slowed campaign execution.

Solution

Designed an autonomous AI agent that retrieves product knowledge through MongoDB Vector Search, generates marketing content using LLMs, creates videos using Google Veo 3, and automatically prepares content for publishing.

Responsibilities

- Designed AI social media workflows.

- Generated marketing captions using LLMs.
- Generated AI-assisted marketing images.
- Created promotional videos using Google Veo 3.
- Built reusable LangGraph orchestration pipelines.
- Automated publishing workflows.

Technologies

- Google AI Studio
 - Google Veo 3
 - OpenAI
 - LangGraph
 - FastAPI
 - AWS Lambda
 - MongoDB Atlas Vector Search
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4. Enterprise AI Chatbot

Overview

Developed a production AI chatbot capable of answering customer inquiries using internal company knowledge instead of relying solely on LLM knowledge. This is part of the Universal Statues Ecommerce site.

Business Problem

Customers required accurate responses based on company-specific information rather than generic AI responses.

Solution

Designed a Retrieval-Augmented Generation (RAG) architecture that retrieves relevant company knowledge before generating responses, significantly improving response accuracy while reducing hallucinations.

Responsibilities

- Designed document ingestion pipelines.

- Built Retrieval-Augmented Generation (RAG) architecture.
- Implemented MongoDB Atlas Vector Search.
- Improved prompt engineering for higher response quality.
- Reduced hallucinations using internal knowledge retrieval.
- Monitored AI traces using LangSmith.

Technologies

- LangChain
 - LangGraph
 - MongoDB Atlas Vector Search
 - VoyageAI
 - OpenAI API
 - LangSmith
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5. Legacy System Modernization

URL: [MKTA FORCE](#)

Overview

Led the modernization of an internal factory operations system by migrating a monolithic application into multiple cloud-native microservices.

Business Problem

The original on-premise monolithic application became unusable during peak production due to hundreds of concurrent users competing for shared computing resources.

Solution

Redesigned the system into department-specific microservices (Casting, Detailing, Assembly, etc.) deployed on AWS Lambda. Migrated the database to MongoDB Atlas to better support the application's document-oriented workloads and distributed architecture.

Responsibilities

- Designed microservice architecture.

- Migrated monolithic applications into independent services.
- Developed REST APIs using FastAPI.
- Deployed workloads to AWS Lambda.
- Implemented AWS SAM deployment pipelines.
- Modernized cloud infrastructure.

Technologies

- FastAPI
 - AWS Lambda
 - AWS SAM
 - API Gateway
 - Docker
 - MongoDB Atlas
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6. Marketplace Integration Platform

Overview

Developed a cloud-native integration platform that synchronizes inventory and orders between Microsoft Dynamics 365 Business Central and multiple online marketplaces.

Business Problem

Managing inventory independently across multiple sales channels created synchronization issues and increased operational overhead.

Solution

Designed a serverless integration platform that automatically synchronizes inventory, orders, pricing, and product information across multiple marketplaces using REST APIs.

Responsibilities

- Integrated Microsoft Dynamics 365 Business Central.
- Built Shopify integrations.
- Built Walmart Marketplace integrations.
- Built eBay integrations.

- Developed inventory synchronization.
- Developed pricing synchronization.
- Built product management APIs.

Technologies

- Python
 - FastAPI
 - REST APIs
 - OAuth
 - AWS Lambda
 - API Gateway
 - MongoDB
 - MySQL
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7. AI Email Generation Engine

Overview

Designed the company's first AI-powered email generation platform, serving as the foundation for the later AI Marketing Agent.

The system remains in production for automated lead nurturing and account creation follow-up emails.

Business Problem

Marketing emails were manually written, leading to inconsistent messaging and slower customer engagement.

Solution

Designed a multi-stage AI workflow that first plans email structure before generating production-ready HTML emails using LLMs.

Responsibilities

- Designed multi-step LangGraph workflows.

- Built AI planning stages.
- Generated HTML emails.
- Improved prompts for layout consistency.
- Automated customer follow-up campaigns.

Technologies

- LangGraph
 - LangChain
 - OpenAI API
 - FastAPI
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8. Internal AI RAG Knowledge Platform

Overview

Developed an internal Retrieval-Augmented Generation platform integrated into the company's HR Information System (HRIS) to assist HR personnel in interpreting company policies and recommending disciplinary actions.

Business Problem

HR personnel manually searched employee handbooks to determine appropriate disciplinary actions, making investigations slower and less consistent.

Solution

Designed a RAG platform that retrieves relevant handbook sections, previous policy references, and disciplinary guidelines while allowing HR personnel to make the final decision.

The AI provides recommendations rather than making decisions autonomously.

Responsibilities

- Built AWS S3 document ingestion pipelines.
- Implemented metadata filtering.
- Designed document chunking strategies.
- Generated vector embeddings.

- Built similarity search using MongoDB Atlas Vector Search.
- Evaluated AI responses using LangSmith.

Technologies

- AWS S3
- LangChain
- LangGraph
- MongoDB Atlas Vector Search
- VoyageAI
- OpenAI API
- LangSmith